



The Robot

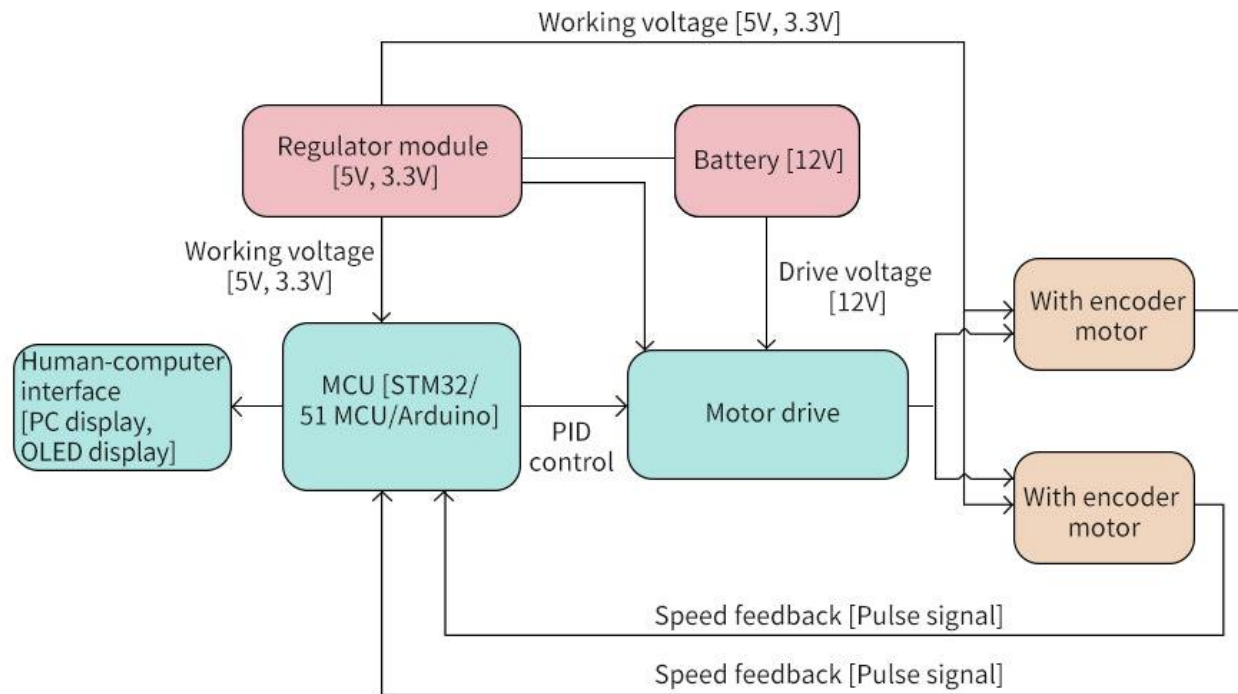
Prof. Muthukumar





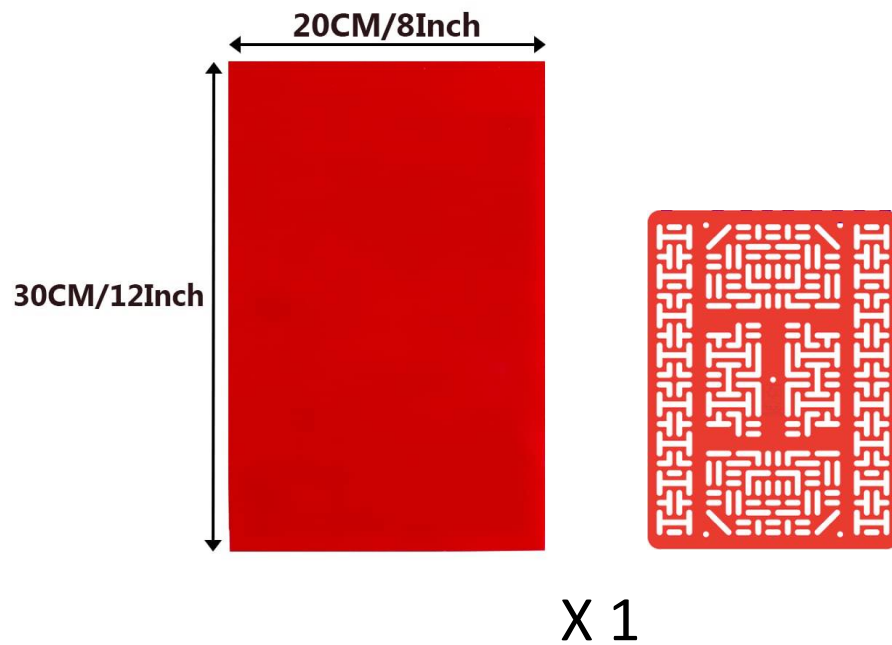
Robot Connection Diagram

Robot car motor control frame diagram





Chassis



Encoder geared motor + motor bracket
+ coupling + rubber tire

12V motor

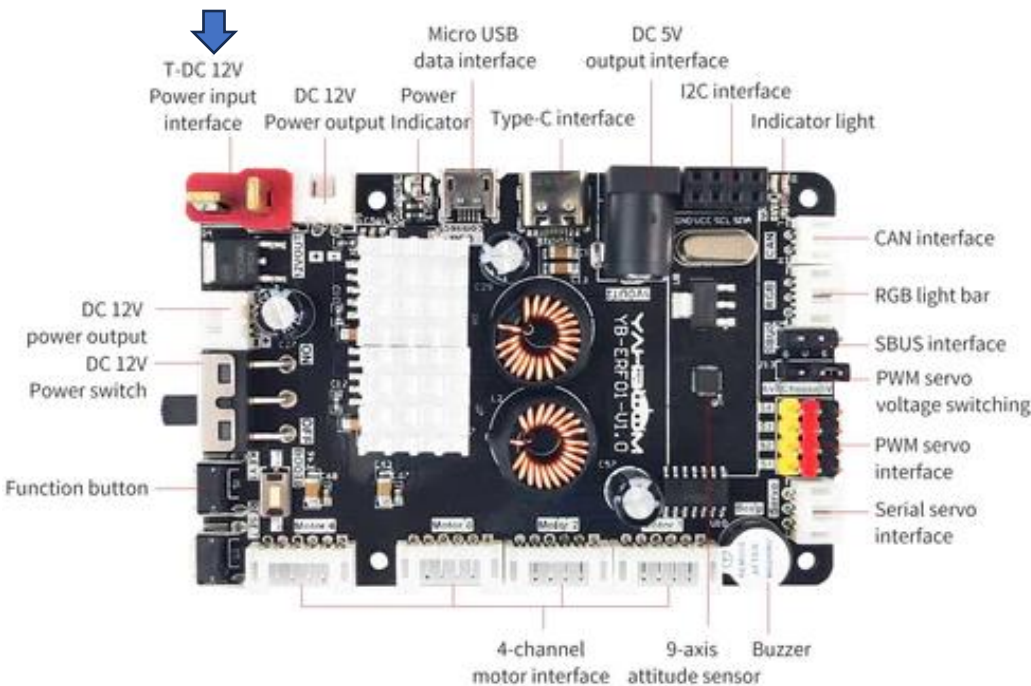


X8 Standoffs





ROS Controller



Basic parameters

On-board MCU model	STM32F103RCT6
Communication baud rate	115200bps
Data interface	micro USB interface
USB to serial port chip	CH340 chip
Data output method	micro USB data interface
Command control	Support serial commands: whistle, light, drive motor, etc.
IMU chip	MPU9250 9-axis attitude sensor
Motor drive model	AM2857 driver chip*4
Motor with encode	Support 4-channel 12V encoder motor
SBUS model aircraft remote control	Support
CAN communication	Support
Other peripheral interfaces	Support PWM servo, serial servo, RGB light bar, OLED display, buzzer
Button	RESET key, KEY1 key, BOOT0 key
Firmware update method	microUSB data interface + MCUISP tool to update firmware
Operating voltage	T-type DC12V input
Stand-by current	About 50mA
Voltage output	DC 12V output interface*2, DC5V output interface*1, Type-C 5V output interface*1
Protection circuit	Servo over-current protection, anti-reverse connection protection
Operating temperature	-40°C~+85°C
Product size	85mm*56mm
Product weight	About 46g





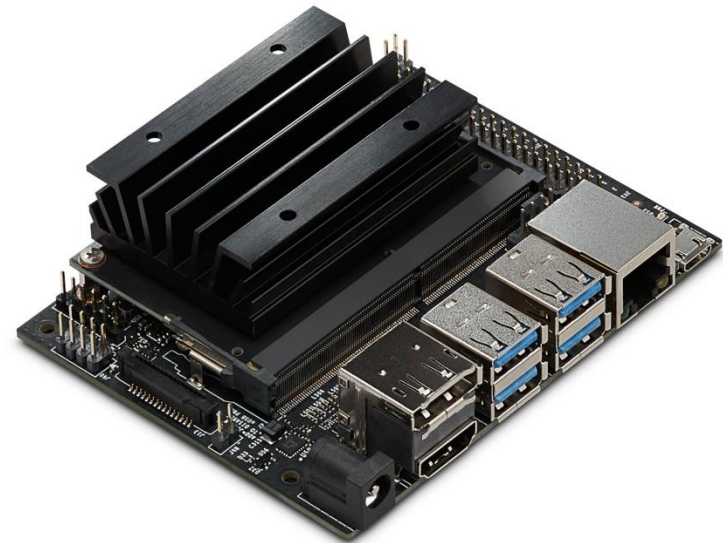
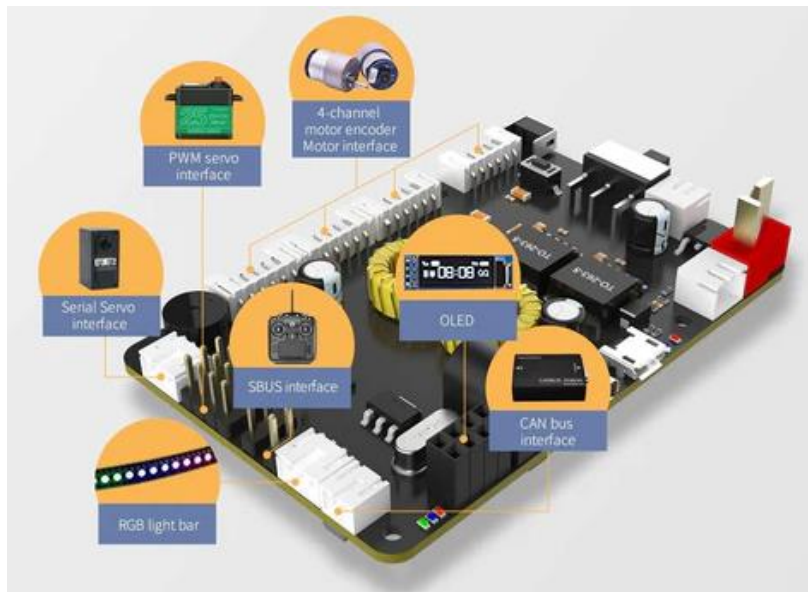
Mobile Wheel Robot Components



X 4
With brackets



Your
responsibility



OLED



RPLIDAR A1



Signal adapter
board



USB data cable



Instruction card



Connection line





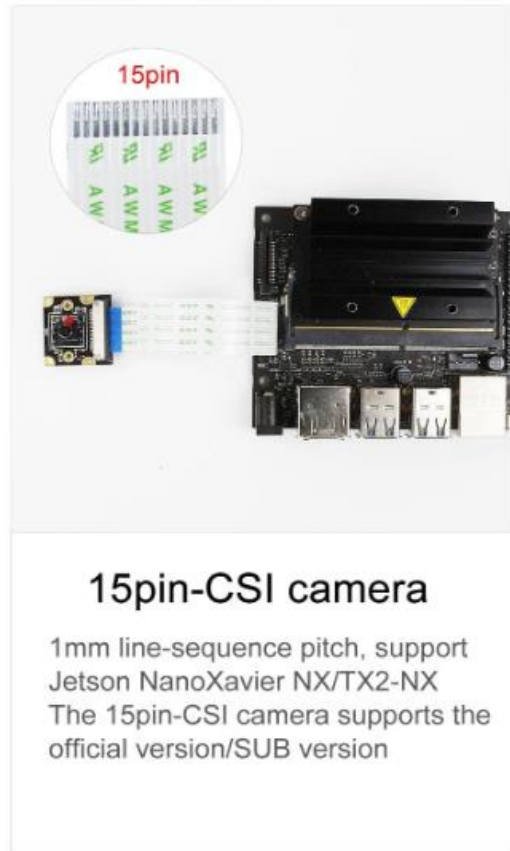
Other Components



JS/PS4 BLE Controller



JS RF Controller



Camera



BLE Dongle



TP-Link
WiFi Dongle



<https://learn.sparkfun.com/tutorials/adding-wifi-to-the-nvidia-jetson/all>

WiFi+BLE Dongle



Jetson Nano WiFi
Dongle

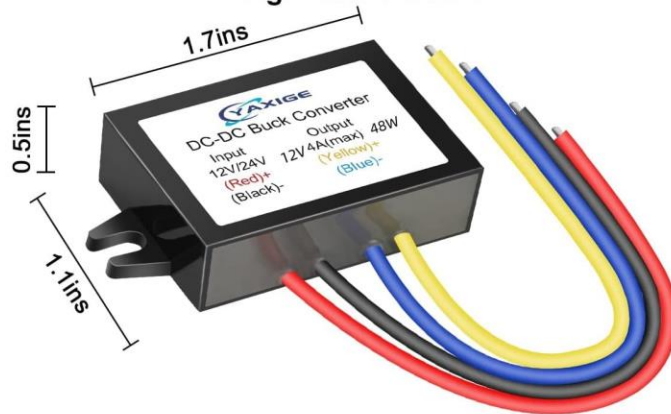




Power Module

12V 24V Power Supply

Light and Portable

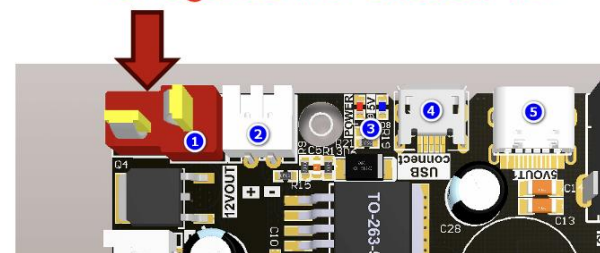


T-Plugs provided

OUTPUT: Female T-Plug
INPUT – Male T-PLUG

input voltage	12V/24V	working temperature	- 20~80C°
output voltage	12 V	ShutDown Current	<10UA
output current	4A(max)	Low voltage protection	sustain
output power	48 W	Overcurrent protection	sustain
conversion efficiency	95%	Wiring length	11MM

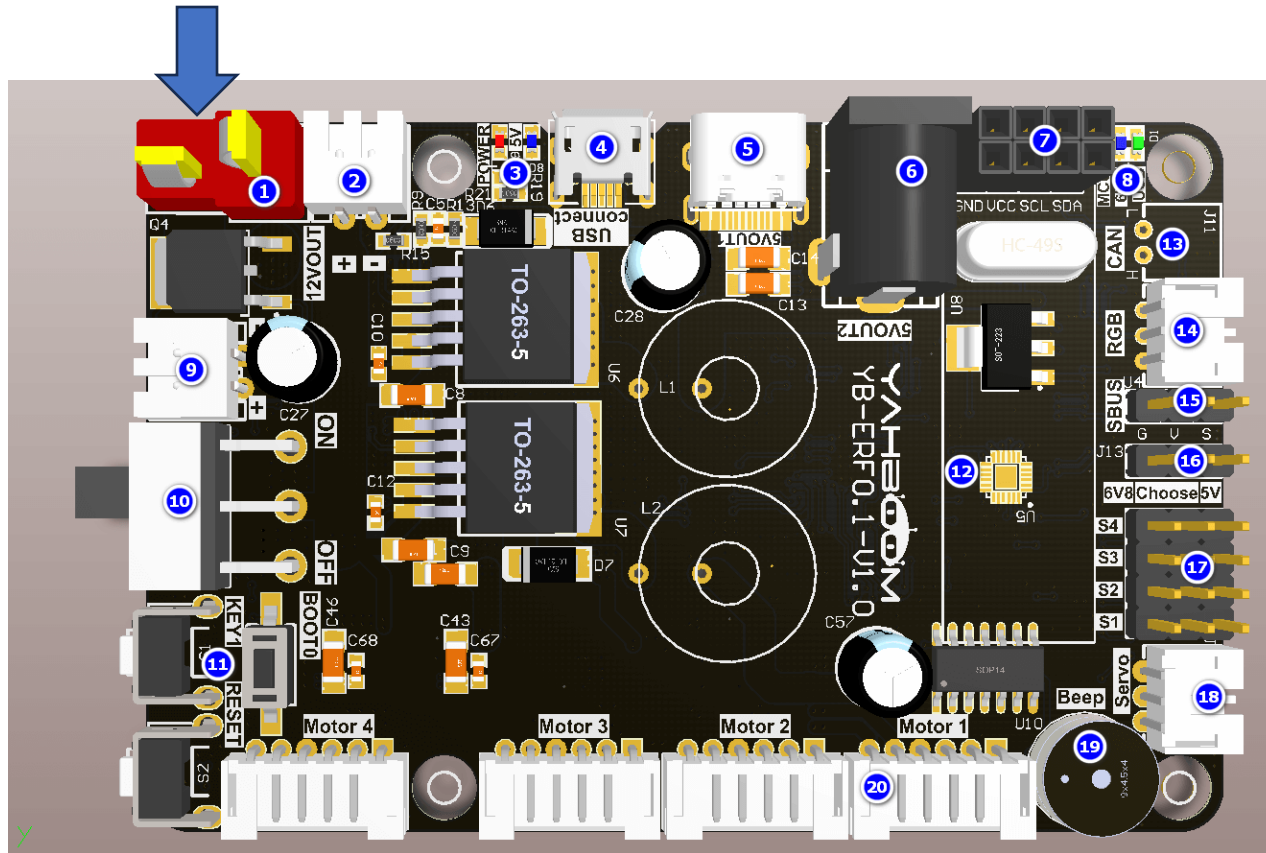
Voltage cannot exceed 12V





Important Power Considerations

Voltage cannot exceed 12V



If using 14.8v Lipo Battery
use the
Provided DC regulator to
step down to 12V



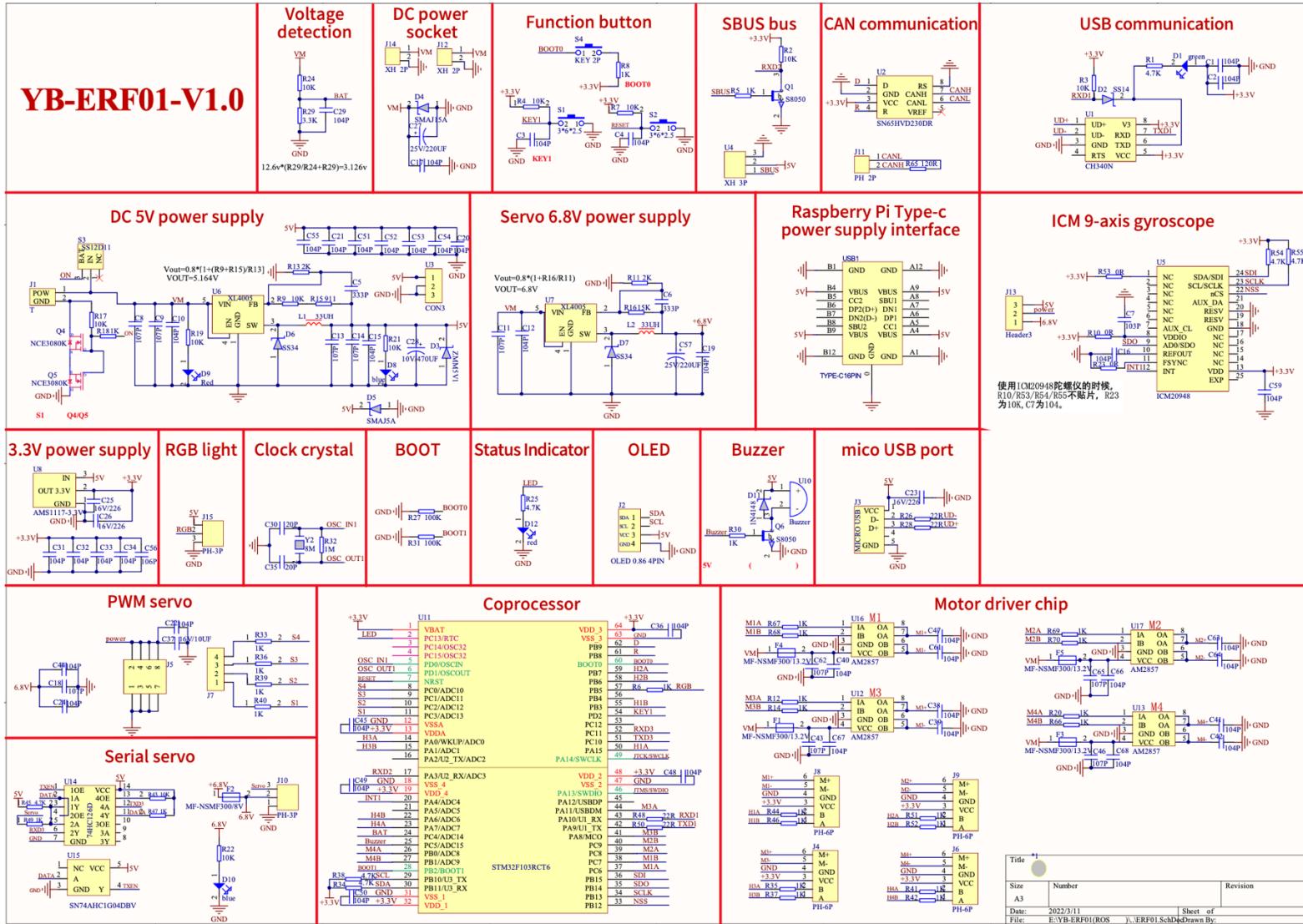
T-Plugs provided

If you damage the ROS controller, you'll need to replace it.
Procurement lead times are long and will cost \$100+.





ROS Controller Schematics





Task for the Weekend

- **Assemble the Robot**

- Component STL/drill templates are uploaded in canvas.
- Be creative and careful.
- Mounts for the Jetson Nano are not provided. Be prepared to make some trips to HD/Lowes!
- Feel free to drill holes, circular and slits (small).

- You'll be required to submit the entire robot as assembled for evaluation for grading. Points awarded for compactness, aesthetics, cleanness of assembly.





ROS robot expansion board

ROS robot expansion board

1. Expansion board STM32 driver ^

0. Precautions

1.1 Expansion Board Introduct...

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1.2 STM32 Development envir...

1.3 Button to control the buzz...

1.4 FreeRTOS application

1.5 Serial communication

1.6 SBUS model aircraft remo...

1.7 CAN bus communication

1.8 RGB colorful light bar

1.9 Timer interrupt to control ...

1.10 Control serial port servos

1.11 Nine-axis attitude sensor...

1.12 Control the forward and r...

1.13 The timer captures the e...

1.14.1 Robot kinematics analy...

1.14.2 Mecanum wheel kinem...

1.15 PID control robot movem...

1.16 OLED character display

<http://www.yahboom.net/study/ROS-Driver-Board>





ROSMASTER X1 repository

- <http://www.yahboom.net/study/ROSMASTER-X1>

ROSMASTER X1

01.Preparation and First Trial



02.Assembly steps



03.Remote control course



04.Hardware course



05.Basic course



06.Linux operating system



07.ROS basic course



08.OpenCV courses



09.Robot control course



10.Depth camera course



11.Lidar course



12.Multi-robot control course



13.Deep learning courses



14.Voice control courses



15.Model training Autopilot

